



Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: April 6, 2026 (Tuesday, ~9:00 a.m.)	Entry: 1
Description	Ransomware attack on a small U.S. healthcare clinic. Employees were unable to access files and software. A ransom note was displayed demanding payment for a decryption key. The attack was carried out via phishing emails with malicious attachments, which installed malware and deployed ransomware that encrypted critical patient data. Business operations were fully shut down.
Tool(s) used	None identified. Incident response and forensic tools would be recommended (e.g., endpoint detection and response platform, email security gateway logs).
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? An organized group of unethical hackers known to target organizations in the healthcare and transportation industries. ● What happened? A ransomware attack encrypted all critical company files, including patient medical records. Employees were locked out of systems and a ransom note was displayed demanding a large sum of money in exchange for the decryption key.

	● When did the incident occur? Tuesday morning at approximately 9:00 a.m. ● Where did the incident happen? At a small U.S. healthcare clinic specializing in primary-care services (internal network and endpoints). ● Why did the incident happen? Attackers sent targeted phishing emails to employees. A malicious email attachment was downloaded, installing malware on the endpoint. Attackers leveraged that access to deploy ransomware and encrypt critical files across the network for financial extortion.
Additional notes	The company was forced to shut down all computer systems and contact multiple external organizations for technical assistance. Attack vector: phishing email → malicious attachment → malware installation → ransomware deployment → file encryption. Key gaps identified: lack of email filtering controls, absence of security awareness training, and no offline backups to restore from. This incident highlights the importance of a tested incident response plan for healthcare organizations handling sensitive patient data.

Date: April 6, 2026	Entry: 2
Description	Hands-on lab: analyzing a network packet capture file using Wireshark. Explored the Wireshark GUI, applied display filters (by IP address, MAC address, UDP/TCP port), and inspected packet data at multiple protocol layers (Frame, Ethernet II, IPv4, TCP). Investigated DNS queries/responses and HTTP web traffic from a sample .pcap file.
Tool(s) used	Wireshark (network protocol analyzer / packet sniffer) — used to open a .pcap

	file, apply display filters, and inspect individual packet layers and payload data.
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? N/A — this was a controlled lab exercise, not a live incident. A sample .pcap file was provided for analysis. ● What happened? A user connected to http://opensource.google.com. DNS queries resolved the hostname to IP 142.250.1.139. TCP connections were established over port 80 (HTTP). ICMP (ping) requests were also captured. Packet-level details were examined: TTL=64, Frame Length=54 bytes, Header Length=20 bytes, Destination=169.254.169.254. ● When did the incident occur? During the lab session; the .pcap file was captured from a Windows-based system during an active web browsing session. ● Where did the incident happen? On the network of the system that generated the traffic. Traffic flowed between the local machine and external servers including a DNS server and the Google-hosted site at 142.250.1.139. ● Why did the incident happen? N/A — lab scenario. The traffic was generated by a user browsing to opensource.google.com. The curl tool was used to make HTTP requests, as confirmed by filtering tcp contains “curl”.
Additional notes	Key filters learned: ip.addr, ip.src, ip.dst (IP-based); eth.addr (MAC-based); udp.port == 53 (DNS); tcp.port == 80 (HTTP); tcp contains “curl” (payload text search). Important insight: DNS traffic uses UDP port 53; the resolved IP for opensource.google.com was 142.250.1.139. MAC address 42:01:ac:15:e0:02 was associated with TCP traffic. Coloring rules in Wireshark help visually classify traffic types quickly (light blue = DNS, green = TCP/HTTP, pink = ICMP). The Frame subtree shows overall packet metadata; Ethernet II shows MAC addresses; IPv4 shows source/destination IPs and internal protocol; TCP

	subtree shows port numbers, sequence numbers, and flags.
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Date: April 6, 2026	Entry: 3
Description	Hands-on lab: capturing and analyzing live network traffic from a Linux virtual machine using tcpdump. Identified available network interfaces, filtered and inspected live traffic, saved packet data to a .pcap file, and filtered the captured data using various tcpdump flags. HTTP traffic to opensource.google.com was captured and examined in both verbose and hex/ASCII output formats.
Tool(s) used	tcpdump (command-line packet analyzer) — used to identify interfaces, capture live traffic, write .pcap files, and filter/read captured packet data. Also used: ifconfig (interface identification), curl (generate HTTP traffic for capture).
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? N/A — controlled lab exercise. The analyst (user: analyst) intentionally generated HTTP traffic using curl to populate the packet capture. ● What happened? Live network traffic was captured on the eth0 interface (IP: 172.17.0.2). A curl request to opensource.google.com generated HTTP traffic on TCP port 80, which was captured into capture.pcap. The capture showed a TCP three-way handshake between 172.17.0.2:46498 and 146.75.38.132:80. Packet data was reviewed in verbose (-v) and hex/ASCII (-X) formats. ● When did the incident occur? During the lab session on a Linux virtual machine (user: analyst already logged in to the terminal).

	● Where did the incident happen? On the eth0 network interface of a Linux virtual machine. Traffic flowed between local IP 172.17.0.2 and external server 146.75.38.132 (opensource.google.com). ● Why did the incident happen? N/A — lab exercise. The analyst deliberately generated and captured HTTP traffic to practice packet analysis. The -nn flag was used to prevent IP/port name resolution, which is security best practice during investigations to avoid alerting threat actors.
Additional notes	Key tcpdump flags learned: -i (specify interface), -v (verbose output), -c (packet count limit), -nn (disable name/port resolution), -w (write to file), -r (read from file), -X (hex + ASCII output), -D (list available interfaces), & (run in background). Important: always use -nn during investigations to avoid inadvertent alerting. The loopback interface (lo: 127.0.0.1) and Ethernet interface (eth0: 172.17.0.2) were both identified via ifconfig. TCP flags observed: [S] = SYN (connection initiation), [S.] = SYN-ACK (server response), [P.] = PSH+ACK (data push). Hex/ASCII output (-X) is especially useful for malware analysis and forensic investigation of packet payloads.

Date: April 6, 2026	Entry: 4
Description	File hash investigation using VirusTotal to determine whether a suspicious file is malicious and to identify associated Indicators of Compromise (IoCs). The file was confirmed malicious — flagged by over 50 security vendors and categorized as Flagpro malware, a tool used by advanced threat actors. IoCs were identified across multiple Pyramid of Pain levels using the Details,

	Relations, and Behavior tabs in VirusTotal.
Tool(s) used	VirusTotal (threat intelligence platform) — used to analyze a file hash, review Detection results, Community Score, security vendor analysis, and sandbox behavior reports (Zenbox, Venus Eye Sandbox, Rising MOVES). Referenced tabs: Detection, Details, Relations, Behavior.
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? Advanced threat actors using Flagpro malware. The malware is associated with command-and-control (C2) infrastructure and input capture capabilities, suggesting an organized, targeted threat group. ● What happened? A suspicious file was submitted for hash analysis. VirusTotal confirmed the file as malicious: over 50 vendors flagged it, multiple vendors categorized it as Flagpro malware. The file communicates with the domain org.misecure.com and IP address 207.148.109.242. It performs input capture (stealing passwords, credit card data) and establishes C2 communication channels. ● When did the incident occur? During a threat intelligence investigation. The file hash was analyzed via VirusTotal to assess risk and gather IoC data. ● Where did the incident happen? On an infected host system. The malware communicates externally with org.misecure.com (resolved to 207.148.109.242) via HTTP requests, indicating outbound C2 traffic from the compromised machine. ● Why did the incident happen? The malware was likely delivered as part of a targeted attack. Its C2 and input capture capabilities suggest the goal was credential theft and persistent remote access over the compromised system.
Additional notes	IoCs identified across the Pyramid of Pain — Hash value (MD5): 287d612e29b71c90aa54947313810a25 (Details tab). Domain name:

	org.misecure.com (Relations tab — flagged as malicious contacted domain). IP address: 207.148.109.242 (Relations tab; also confirmed in DNS Resolutions under Behavior tab, Zenbox sandbox). Network/host artifact: HTTP requests to org.misecure.com observed in Network Communications (Venus Eye Sandbox and Rising MOVES sandbox, Behavior tab). Tool: Input capture — listed under Collection in Behavior tab (Zenbox); used to steal passwords and sensitive user input. TTP: Command and Control (C2) — listed as a tactic in Behavior tab (Zenbox); enables attacker to communicate with and control the infected system remotely. Key takeaway: Higher-level IoCs on the Pyramid of Pain (TTPs, tools) are harder for attackers to change and more valuable for long-term detection. VirusTotal’s Community Score and multi-vendor consensus are strong signals when triaging unknown files.
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Date: April 6, 2026 (Incident date: July 20, 2022, 09:30 AM)	Entry: 5
Description	SOC Tier-1 alert triage — Ticket A-2703 (Severity: Medium). An employee at Inergy received a phishing email disguised as a job application from “Def Communications.” The email contained a password-protected malicious executable (bfsvc.exe) which was downloaded and opened on the affected machine. The file hash was confirmed malicious. Multiple red flags identified: mismatched sender identity, suspicious domain (.su TLD), grammatical errors,

	and executable disguised as a resume. Ticket escalated to Level-2 SOC analyst.
Tool(s) used	SIEM / Alert ticketing system (ticket A-2703). VirusTotal (previously used to confirm file hash 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b as malicious). Email header analysis (manual inspection of sender address, IP, and metadata).
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? An external threat actor posing as a job applicant named “Clyde West” from “Def Communications.” The sending address (76tguyhh6tgftrt7tg.su, IP: 114.114.114.114) is inconsistent with the claimed identity and uses a .su (Soviet Union) TLD — a known indicator of malicious infrastructure. ● What happened? A phishing email was sent to hr@inergy.com impersonating a job applicant. The email contained a password-protected attachment (bfsvc.exe) with the password provided in the email body (paradise10789). The HR employee downloaded and opened the file, executing a known malicious binary on the machine. Hash confirmed: 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b. ● When did the incident occur? Wednesday, July 20, 2022 at 09:30:14 AM. ● Where did the incident happen? On the affected employee’s machine at Inergy (recipient IP: 176.157.125.93). The malicious file was executed locally after being downloaded from the phishing email. ● Why did the incident happen? The attacker used a spear-phishing lure (fake job application) to socially engineer an HR employee into opening a malicious executable. The password protection on the attachment

	was used to bypass email security scanning tools that cannot inspect password-protected files.
Additional notes	Red flags identified: (1) Sender email domain mismatch — display name “Def Communications” does not match sending address 76tguyhh6tgftrt7tg.su. (2) .su TLD is associated with high-risk or malicious infrastructure. (3) Grammatical errors in subject line (“Egnieer” instead of “Engineer”) and body (“I am writing for to express”). (4) Executable file (.exe) disguised as a resume — legitimate resumes are submitted as .pdf or .docx. (5) Password provided inline — a social engineering tactic to bypass attachment scanning. (6) File hash 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b confirmed malicious via prior VirusTotal analysis. Escalation rationale: medium severity + confirmed malicious execution on endpoint warrants Level-2 investigation for containment, host forensics, and potential lateral movement assessment. Recommended follow-up: isolate affected machine, reset credentials for the HR user, scan environment for additional indicators of this hash.

Date: April 6, 2026	Entry: 6
Description	Hands-on lab: configuring and running Suricata as a network intrusion detection system (IDS). Examined the anatomy of a custom Suricata rule, triggered alerts against a sample.pcap file, and analyzed output logs in both fast.log and eve.json formats. Used the jq command-line tool to parse and filter JSON log data. Key concepts covered: rule structure (action, header, rule options), log formats, flow_id correlation, and network traffic monitoring.

Tool(s) used	Suricata (open-source IDS/IPS/network analysis tool) — used to process a packet capture file against custom rules and generate alert logs. jq (command-line JSON processor) — used to parse and filter eve.json output. cat / ls — used to view log files. Linux Bash shell (user: analyst).
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? N/A — controlled lab exercise. Alerts were intentionally triggered by running Suricata against a sample.pcap file to simulate network monitoring. ● What happened? Suricata processed sample.pcap using a custom rule that detected outbound HTTP GET requests from the home network (\$HOME_NET: 172.21.224.0/20) to the external network. Two alerts were generated in fast.log: traffic from 172.21.224.2 to 142.250.1.139:80. The eve.json file contained detailed telemetry including timestamps, flow IDs, protocols, and destination IPs (142.250.1.139 and 142.250.1.102). Alert severity was 3; alert signature was “GET on wire.” ● When did the incident occur? November 23, 2022 (timestamps from sample.pcap: 12:38:34 and 12:38:58 UTC). ● Where did the incident happen? On the monitored network interface. Traffic originated from internal host 172.21.224.2 and was destined for external IPs 142.250.1.139 and 142.250.1.102 over TCP port 80 (HTTP). ● Why did the incident happen? N/A — lab scenario. The custom rule was designed to detect outbound HTTP GET requests, which is a common pattern to monitor for potential data exfiltration or unauthorized web communication from internal hosts.
Additional notes	Suricata rule anatomy: Action (alert/drop/pass/reject) — Header (protocol, source IP:port, direction →, destination IP:port) — Rule Options (msg, flow, content, http_method, sid, rev). Rule processing order: pass → drop → reject → alert. Key rule options: msg (alert text), flow:established,to_server (packet direction), content:“GET” + http_method (payload match), sid (unique rule ID),

	rev (rule version). Log formats: fast.log is a deprecated quick-check format; eve.json is the standard format for incident response and threat hunting. jq extraction syntax used: <code>jq -c "[.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]"</code> to pull key fields. flow_id (16-digit number) allows correlation of all packets belonging to the same network flow. The <code>-k none</code> flag disables checksum validation when replaying .pcap files. \$HOME_NET and \$EXTERNAL_NET are Suricata variables defined in suricata.yaml. Suricata can run as IDS (alert only) or IPS (alert + drop/reject traffic).
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Need another journal entry template?

If you want to add more journal entries, please copy one of the tables above and paste it into the template to use for future entries.

Reflections/Notes: Record additional notes.
